STUDENT NUMBER:

APSC 171 - Term Test 2
Tuesday November 5, 2024

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Last Name (BLOCK LETTERS): _____

First Name: _____

Student #: _____

- This is a 2 hour test.
- No textbooks, notes, or aids other than your calculator (Casio 991 series) are allowed.
- For full marks, you must show all your work and explain how you arrived at your answers, unless explicitly told to do otherwise.
- Wherever appropriate, **include units in your answers**.
- When drawing graphs, **add labels and scales on all axes**.
- Correct final answers with incorrect support may be given no credit.
- While waiting to start, fill in your name and student number above.
- If you need more room, there are blank pages at the end of the test. **If you use these pages, you must provide clear directions on the original page**, e.g. "Continued on page 10".
- Only answers written in this booklet will be graded; all scrap paper from the test will be discarded.

Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Total
6	12	6	4	5	7	6	7	7	60

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1. The power used by a motor can be measured in watts; this is also a measure of the **rate** of energy usage, because (1 watt) = 1 Joule/second.

   Below is a table showing the power used by an electric motor driving a small cart at high speed around a long-distance course.

Time (s)	0	15	30	45	60	75	90
Power (W)	45	42	46	52	39	16	0

values for RIGHT(6)

   (a) Use a right sum with 6 intervals, i.e. RIGHT(6), to estimate the total energy (in Joules) used by the motor during the cart's 90 second run.
   Reminder from the front page: include units in all answers when they are available. [/3]

RIGHT(6) = $\sum$ (power)(time interval)

W · s = J.

= (42 + 46 + 52 + 39 + 16 + 0)(15)

= 2925 J

   (b) Use a trapezoidal sum with 6 intervals, i.e. TRAP(6), to estimate the total energy (in Joules) used by the motor during the cart's 90 second run. [/2]

TRAP(6) = $\sum$ (avg power)(time interval)

= $\left(\frac{45+42}{2} + \frac{42+46}{2} + \cdots + \frac{16+0}{2}\right)(15)$

= 3262.5 J

   (c) In the experiment above, a single fresh battery was used to power the motor, and by the end of the 90 seconds, the battery's energy was all discharged. Based on your energy usage estimate from parts (a) and (b), which battery type, AAA, AA, C or D, was used to power the motor?
   Note that total battery energy is usually reported in mAh (milli-amp hours); AAA, AA, C and D batteries are 1.5 V batteries, and for them 1 mAh = 5.4 J. [/1]

   - AAA: 600-1000 mAh
   - AA: 2000-3000 mAh
   - C: 5000-7000 mAh
   - D: 13000-17000 mAh

The battery was a AAA battery.

RIGHT(6) = 2925 J · $\left(\frac{1\text{ mAh}}{5.4\text{ J}}\right)$ ≈ 542 mAh

TRAP(6) = 3262.5 J · $\left(\frac{1\text{ mAh}}{5.4\text{ J}}\right)$ ≈ 604 mAh

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2. Evaluate the following definite and indefinite integrals. As always, show your work.

(a) $\int xe^{-2x}\,dx$ [/4]

by parts: let $u = x$ and $\int dv = \int e^{-2x}\,dx$

$\downarrow d/dx$

$du = 1 \cdot dx$ $\quad\leftarrow\quad$ $v = \frac{e^{-2x}}{-2}$

$= (x)\left(\frac{e^{-2x}}{-2}\right) - \int \left(\frac{e^{-2x}}{-2}\right) dx$

tidy $= -\frac{xe^{-2x}}{2} + \frac{1}{2}\int e^{-2x}\,dx$

integrate $= -\frac{xe^{-2x}}{2} + \frac{1}{2}\frac{e^{-2x}}{-2} + C$ $\left(\text{or tidying again} = -\frac{xe^{-2x}}{2} - \frac{e^{-2x}}{4} + C\right)$

(b) $\int x^2 \sin(x^3)\,dx$ [/4]

Substitution: let $w = x^3$

so $\frac{dw}{dx} = 3x^2$ or $\frac{1}{3x^2}\,dw = dx$

$= \int x^2 \cdot \sin(w)\left(\frac{1}{3x^2}\,dw\right)$

$= \frac{1}{3}\int \sin(w)\,dw$

$= 1/3\,(-\cos(w)) + C$

$= -\frac{1}{3}\cos(x^3) + C.$

(c) $\int_0^{0.5} \frac{1}{\sqrt{1-x^2}}\,dx$ [/4]

$= \arcsin(x)\Big|_{x=0}^{x=0.5}$

$= \arcsin(0.5) - \arcsin(0)$

$= \pi/6 - 0$

$= \pi/6$

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3. Consider the integral $\int x \ln(\sqrt{x})\, dx$. $= \int x \ln(x^{1/2})\, dx = I$

(a) Evaluate the indefinite integral. [/5]

Using by parts as-is: let $u = \ln(x^{1/2})$ and $\int du = \int x\, dx$

$\downarrow d/dx$

$du = \frac{1}{x^{1/2}} \cdot \frac{1}{2} x^{-1/2}\, dx$ $\quad u = \frac{x^2}{2}$

or $du = \frac{1}{2x}\, dx$

so $I = \ln(\sqrt{x}) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \left(\frac{1}{2x}\, dx\right)$

tidy $= \frac{1}{2}\ln(\sqrt{x})\, x^2 - \frac{1}{4}\int x\, dx$

integrate $= \frac{1}{2}\ln(\sqrt{x})(x^2) - \frac{1}{4}\frac{x^2}{2} + C$

$= \frac{1}{2}\ln(\sqrt{x})(x^2) - \frac{x^2}{8} + C.$

(b) Determine by differentiating whether or not your answer to part (a) is correct. State your conclusion clearly for the grader. [/1]

$\frac{d}{dx}\left[\frac{1}{2}\ln(\sqrt{x})(x^2) - \frac{x^2}{4} + C\right]$

$= \frac{1}{2}\frac{1}{\sqrt{x}} \cdot \frac{1}{2} x^{-1/2} \cdot x^2 + \frac{1}{2}\ln(\sqrt{x})(2x) - \frac{2x}{8}$

$= \frac{1}{4x} + x\ln(\sqrt{x}) - \frac{x}{4}$

$= x\ln(\sqrt{x})$, the original integrand.

Part (a) is correct.

Q 3: Alternative Approach

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3. Consider the integral $\int x \ln(\sqrt{x})\, dx$. $= \int x \ln(x^{1/2})\, dx = I$

(a) Evaluate the indefinite integral. [/5]

Noting $\ln(x^{1/2}) = \frac{1}{2}\ln(x)$ using ln power rules,

$I = \frac{1}{2}\int x \ln(x)\, dx$ By parts,

let $u = \ln(x)$ $\int dv = \int x\, dx$

$\downarrow d/dx$ $\rightarrow$

$du = \frac{1}{x}dx$ $\leftarrow$ $v = \frac{x^2}{2}$

$= \frac{1}{2}\left[\ln(x)\cdot\frac{x^2}{2} - \int \left(\frac{x^2}{2}\right)\left(\frac{1}{x}dx\right)\right]$

tidy $= \frac{1}{2}\left[\frac{x^2}{2}\ln(x) - \frac{1}{2}\int x\, dx \right]$

$= \frac{1}{2}\left[\frac{x^2}{2}\ln(x) - \frac{1}{2}\frac{x^2}{2} \right] + C$

$= \frac{1}{2}\left[\frac{x^2}{2}\ln(x) - \frac{x^2}{4} \right] + C$

or $= \frac{x^2}{4}\ln(x) - \frac{x^2}{8} + C$

(b) Determine by differentiating whether or not your answer to part (a) is correct. State your conclusion clearly for the grader. [/1]

$\frac{d}{dx}\left(\frac{x^2}{4}\ln(x) - \frac{x^2}{4} + C\right)$

$= \frac{x}{2}\ln(x) - \frac{x^2}{4}\cdot\frac{1}{x} - \frac{x}{4}$

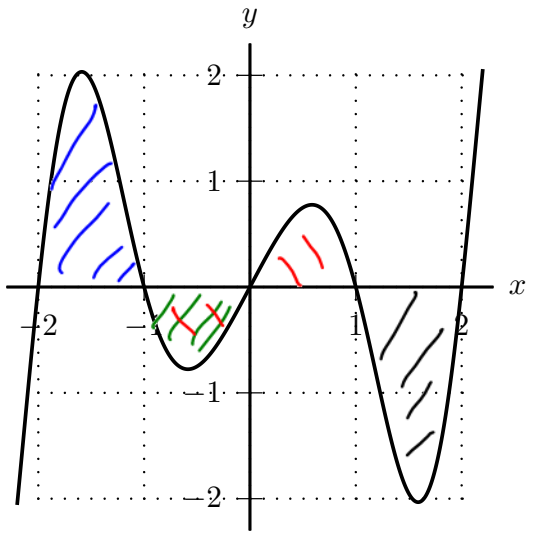
$= \frac{1}{2}x\ln(x)$ but $\frac{1}{2}\ln(x) = \ln(x^{1/2}) = \ln(\sqrt{x})$

$= x\ln(\sqrt{x})$, the original integrand.

Part (a) is correct.

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4. Consider the function $f(x)$ graphed at right.



We are interested in ranking the value of the following definite integrals.

A. $\int_{-2}^{-1} f(x)\,dx$. ≈ 1.5

B. $\int_{-1}^{0} f(x)\,dx$. ≈ -0.7

C. $\int_{-1}^{1} f(x)\,dx$. $= 0$ from symmetry

D. $\int_{1}^{2} f(x)\,dx$. ≈ -1.5

Put the capital letters representing the value of the associated integral into the inequality sequence below. For this question no explanation or calculations are necessary.
Note there are 6 slots but only 4 letters/integrals, so two spaces will have to be left blank. [/4]

____ ≤ D ≤ -1 ≤ B ≤ C ≤ 1 ≤ A ≤ ____

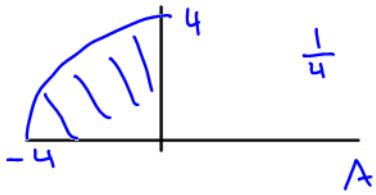
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5. (a) Without using an anti-derivative, determine the **exact** value of the integral $I = \int_{-4}^{0} (\sqrt{(16-x^2)}-2)\,dx$.

Hint: a sketch will be essential. Decimal answers, e.g. 4.1234, will not be accepted for this question. [/3]

$I = \int_{-4}^{0} \sqrt{16-x^2}\,dx - 2\int_{-4}^{0} 1\,dx$

I_1 I_2

I_1: 

$\frac{1}{4}$ of a circle area, $R = 4$:

$A = \frac{1}{4}\pi(4)^2 = 4\pi$

I_2 = rectangle, 4×1:

$I = 4\pi - 2(4) = \boxed{4\pi - 8}$

(b) Compute the exact value of the related integral $\int_{-4}^{0} (5\sqrt{(16-x^2)}-2)\,dx$. [/2]

$= 5\int_{-4}^{0} \sqrt{16-x^2}\,dx - 2\int_{-4}^{0} 1\,dx$

$= 4\pi$ from (a); $= -8$ from A

$= 20\pi - 8$

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6. An LCD monitor has a 4:3 ratio of width to height. The sale price p of a monitor (in dollars) is determined by the length of its diagonal length (in inches):

$$p(d) = 8.5\ d.$$

The manufacturing cost m, however, is related to the area a of the monitor (in square inches):

$$m(a) = \frac{a^2}{8000}.$$

What size of monitor makes the most profit (i.e., maximizes $p - m$)? **As your final answer, report the length of the optimal design's diagonal length** d.
Note: you do **not** need to use the first or second derivative test for this problem to confirm the critical point type. [/7]

$d = 5x$, $a = (4x)(3x) = 12x^2$



Let side lengths be $4x$ and $3x$

$\Rightarrow$ diagonal is $\sqrt{(4x)^2 + (3x)^2}$

$= \sqrt{25x^2} = 5x$

Profit$(x) = p - m$

$= 8.5(5x) - \frac{(12x^2)^2}{8000}$

tidy $= 42.5x - \frac{144x^4}{8000}$ or $\frac{9}{500}$ or 0.018

Find critical points of Profit:

$\frac{dP}{dx} = 42.5 - 0.018(4)x^3$ Set $= 0$

$x^3 = \frac{42.5}{0.072} \approx 590.28$

So $x = \sqrt[3]{590.28} = 8.39$ inches

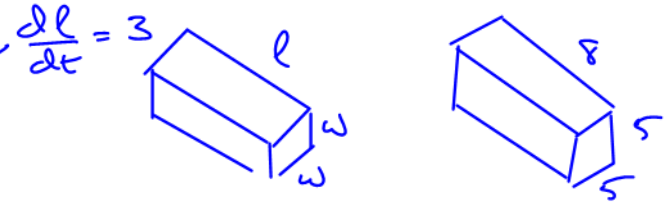
Diagonal is $d = 5x \approx 41.9$ inches

Optimal diagonal d: 41.9 inches.

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7. A compressed dry sponge is initially 8 cm in length, and 5 cm in both width and height. It is dropped in water and begins to expand. You note that:

   - the width and height remain equal to each other;
   - the length increases at a rate of 3 cm/min; and
   - the volume increases at a rate of 120 cm^3/min.



Initial

dV/dt = 120

(a) At the moment the sponge is dropped in the water, how fast is its width increasing? [/4]

Since $V = l \cdot w^2$ for the sponge,

$$\frac{d}{dt}(V) = \frac{d}{dt}(l w^2)$$

$$\frac{dV}{dt} = \frac{dl}{dt} \cdot w^2 + l \cdot 2w \frac{dw}{dt}$$

$\frac{dV}{dt} = 120$, $\frac{dl}{dt} = 3$

Initially, $w = 5$, $l = 8$

$$120 = 3w^2 + l \cdot 2w \frac{dw}{dt}$$

$$120 = \underbrace{3 \cdot 5^2}_{=75} + \underbrace{(8)(2 \cdot 5)}_{=80} \frac{dw}{dt}$$

so $80 \frac{dw}{dt} = 120 - 75 = 45$, or $\frac{dw}{dt} = \frac{45}{80} = 0.5625$ cm/min.

(b) At what rate is the **surface area** of the sponge increasing at this same moment? [/2]

$$SA = 2w^2 + 4w \cdot l$$

so $$\frac{d(SA)}{dt} = 4w \frac{dw}{dt} + 4\left[\frac{dw}{dt} \cdot l + w \cdot \frac{dl}{dt}\right]$$

all the values from earlier

$$= 4(5) \cdot (0.5625) + 4\left[(0.5625)(8) + (5)(3)\right]$$

$$= 11.25 + 4(19.5)$$

$$= 89.25 \text{ cm}^2/\text{min}.$$

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8. The relation $(x-y)^2+6x-10y+29=0$ ① defines a parabola.

(a) Find the (x,y) coordinates of point(s) where the parabola has a horizontal tangent line. [/3]

Implicit Derivative: $\frac{d}{dx}\left[(x-y)^2+6x-10y+29\right]=\frac{d}{dx}[0]$

$2(x-y)\left(1-\frac{dy}{dx}\right)+6-10\frac{dy}{dx}=0$

options: • solve for $\frac{dy}{dx}$, or
• set $\frac{dy}{dx}=0$ now for horiz tgt lines.
We will solve for $\frac{dy}{dx}$.

Grouping terms: $-2(x-y)\frac{dy}{dx}-10\frac{dy}{dx}=-2(x-y)-6$

Divide by -2: $\frac{dy}{dx}(x-y+5)=x-y+3$

$\frac{dy}{dx}=\frac{x-y+3}{x-y+5}$

Set $\frac{dy}{dx}=0$: $0=x-y+3 \rightarrow x-y=-3$ ② or $x=y-3$

② → ①

$(-3)^2+6(y-3)-10y+29=0$

$-4y+9-18+29=0$

$4y=20$

$y=5$

↳ $x=y-3=2$ $\boxed{(2,5)}$ is the only point

(b) The point (5,6) lies on the parabola. Find the formula for the tangent line to the parabola at the point (5, 6). [/3]

At $(5,6)$, $\frac{dy}{dx}=\frac{x-y+3}{x-y+5}=\frac{5-6+3}{5-6+5}=\frac{2}{4}=\frac{1}{2}$

Tgt line is $y=\frac{1}{2}(x-5)+6$

(c) Using your answer to part (b), estimate the value of y on the parabola at $x=4.95$. [/1]

@ $x=4.95$, $y\approx\frac{1}{2}(4.95-5)+6$

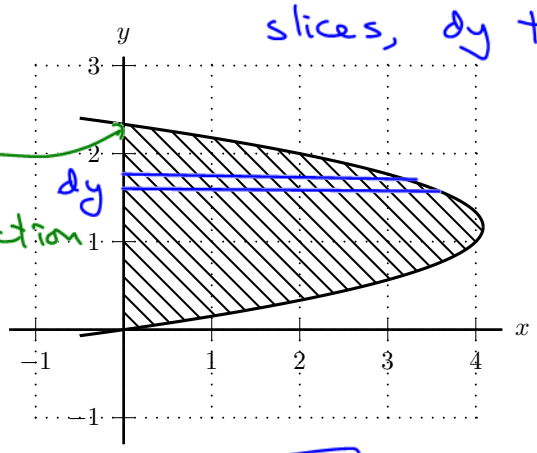
$=5.975$

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9. The following questions are unrelated.

(a) Find the area of shaded region at right, bounded between the vertical axis and the relation $x = 7y - 3y^2$. [/4]

Taking horizontal slices, dy thick



Finding this intersection point:

$x = 0$ and $x = 7y - 3y^2$

$0 = 7y - 3y^2$

$0 = y(7 - 3y) \rightarrow y = 0, \boxed{y = \frac{7}{3}}$

$$\text{Area} = \int_{y=0}^{y=7/3} (7y - 3y^2)\, dy$$

$$= \frac{7y^2}{2} - \frac{3y^3}{3} \Big|_{y=0}^{y=7/3} = \left[\left(\frac{7}{2}\right)\left(\frac{7}{3}\right)^2 - \left(\frac{7}{3}\right)^3\right] - [0 - 0]$$

$$= \frac{343}{54} \approx 6.35$$

(b) Indicate whether the following statements are true or false by circling the appropriate answer. No explanation is required. [/3]

When computing an indefinite integral like $\int f(x)\,dx$, the $+C$ term in the anti-derivative is optional because it will cancel later in the calculation. T / (F)

only true for definite integrals

Consider a function $g(x)$ with a critical point $x = a$ where $g'(a) = 0$. The second derivative test indicates whether a critical point is a local max if $g''(a) < 0$, local min if $g''(a) < 0$, or undetermined if $g''(a) = 0$. T / (F)

NOTE: All students were given credit for this question, regardless of their answer.

Consider a function $f(x)$ that is always increasing and concave down. On any interval, the TRAP sum approximation to the integral of $f(x)$ will always under-estimate the corresponding exact integral value of $f(x)$. (T) / F



TRAP under estimates

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Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

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